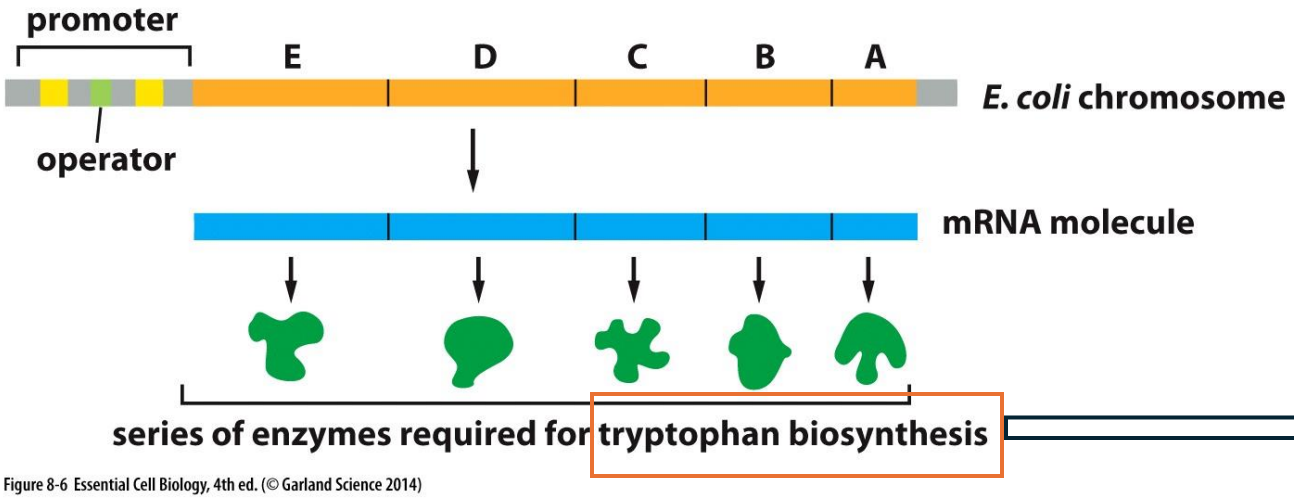
Prokaryotic Transcription

© 2016 Pearson Education, Inc.

Prokaryotic gene expression



When needed, all related proteins are synthesized at the same time from a single mRNA. Saves space and increase feasibility of txn.

Figure 8-6 Essential Cell Biology, 4th ed. (© Garland Science 2014)

- **Polycistronic mRNA:** in bacteria, they encode several polypeptides, usually with related functions in the cell
- **Operon:** The clusters of genes that give rise to polycistronic mRNAs are single transcription units called "operons"

Operator regions are bound by repressor proteins, based on the availability of nutrients

Nutrients bind to repressors, change their 3D shape and activity.

Regulatory regions, they are not transcribed (not genes!)

They are bound by proteins functioning in txn

RNA Polymerase cannot bind to promoter region

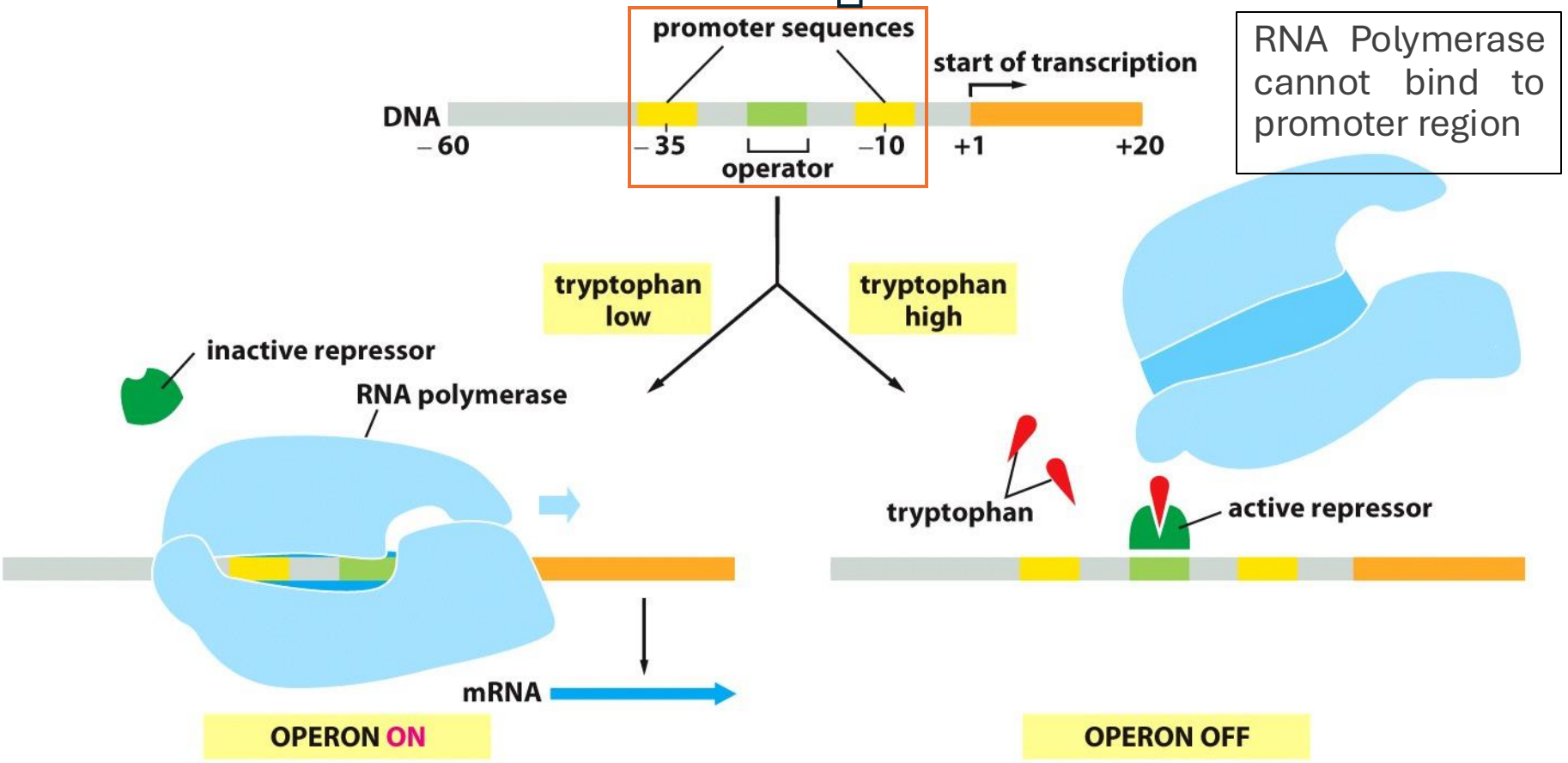


Figure 8-7 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Binding of RNA Polymerase to a Promoter Sequence

- RNA polymerase binds to a DNA promoter site, a sequence of several dozen base pairs that determines where RNA synthesis will start
- The terms **upstream** and **downstream** refer to sequences located toward the 5' or 3' end of the transcription unit, respectively
- **The promoter is upstream of the transcribed sequence**
- Transcription start site is the reference point "+1"

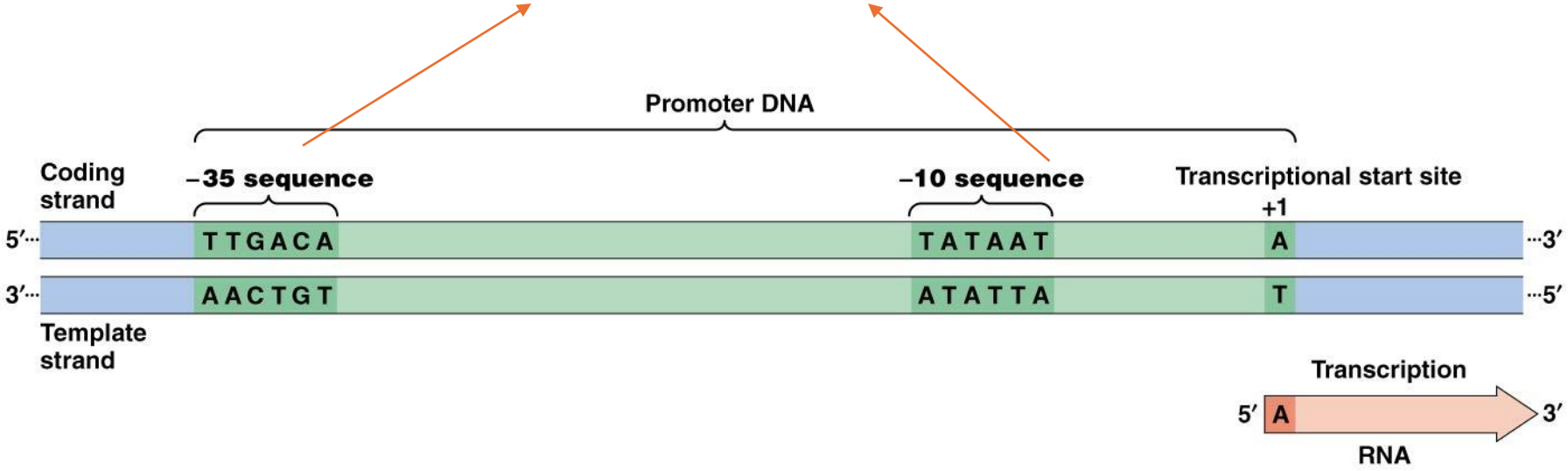
© 2016 Pearson Education, Inc.

Essential Sequences in a Typical Bacterial Promoter

- For landing path, a specific signal/sequence is required for RNA pol binding:
- The transcription start site is almost always a purine and usually an adenine
- About 10 bp upstream of the start site is the sequence TATAAT, called the *–10 sequence* or the *Pribnow box*
- At or near the –35 position is the sequence TTGACA, called the *–35 sequence*
- These sequences are evolutionarily conserved.

© 2016 Pearson Education, Inc.

The proteins that bind to those regions enable RNA polymerase binding for initiation of transcription.



- The transcription start site is almost always a purine and usually an adenine
- About 10 bp upstream of the start site is the sequence TATAAT, called the *–10 sequence* or the *Pribnow box*
- At or near the –35 position is the sequence TTGACA, called the *–35 sequence*

© 2016 Pearson Education, Inc.

RNA Polymerase Subunits

- Bacterial cells have a single kind of RNA polymerase (unlike eukaryotes) that synthesizes all major classes of RNA
- RNA polymerase is a large protein consisting of two α subunits, two β subunits, and a dissociable subunit called the **sigma** (**σ**) **factor**

© 2016 Pearson Education, Inc.

How Is the Information Content in DNA Transcribed to Produce RNA?

- **In, prokaryotes: Sigma factors** and **transcription factors** are proteins that bind to DNA sequences and to RNA polymerase.
- *They help direct the polymerase onto the promoter, and help determine which genes are expressed at particular times.*
- Multiple sigma factors exist for particular conditions! (remember our discussion from last week, on how the promoter defines the specificity of transcription)

© 2016 Pearson Education, Inc.

Prokaryotic transcription

- Sigma factors (yellow circle) recognize the start sites of genes and direct RNA polymerase there.
- The transcription start sites of genes are called **promoter**s.
- Promoter region contain sequences that are recognized by sigma factors that are required for a particular condition.
- The DNA template unzips/melts and sigma factor leaves. RNA polymerase synthesizes the mRNA. Templatre strand resides in the active site.
- In prokaryotes, transcription termination happens in specific DNA sites called **terminator**s.

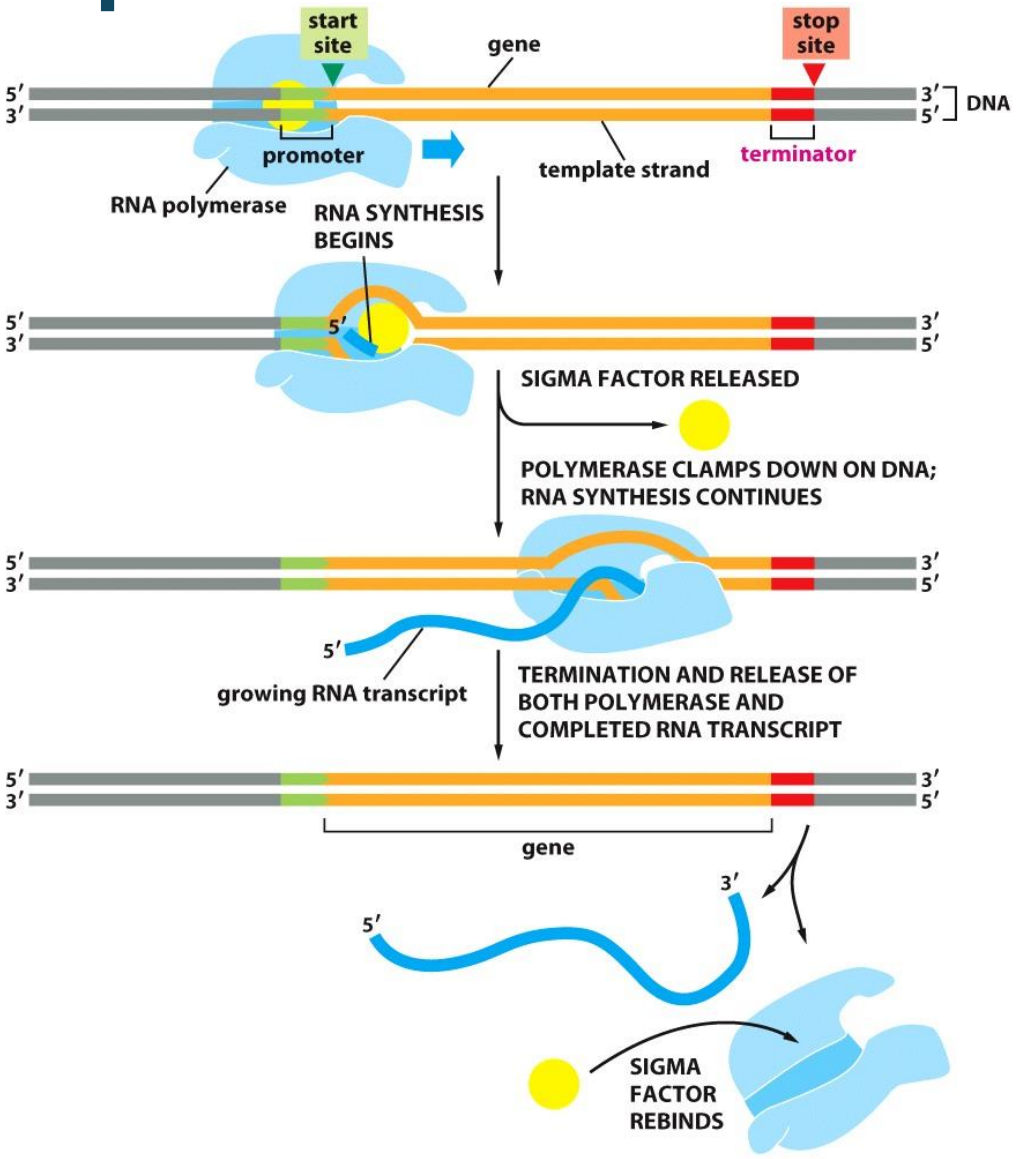


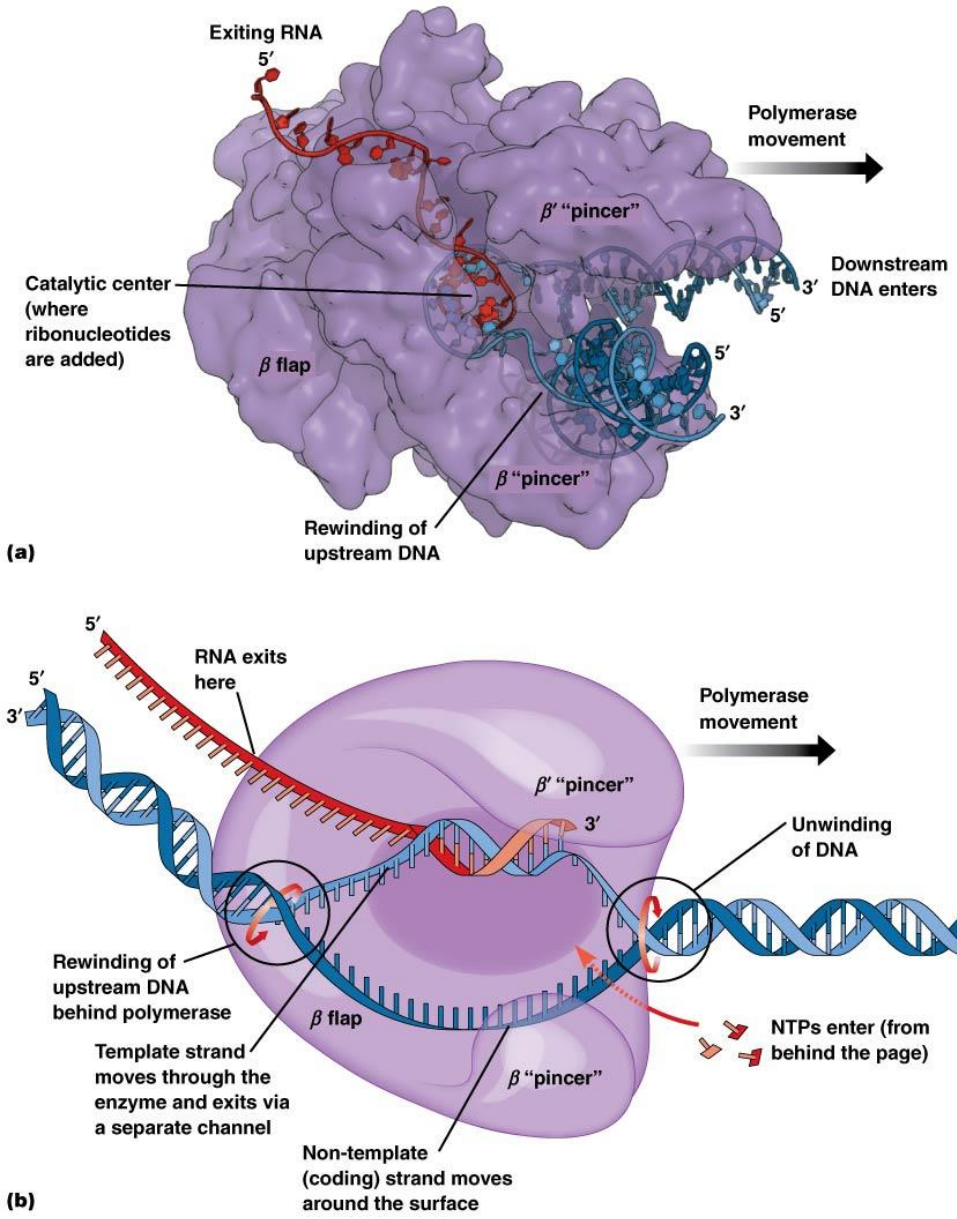
Figure 7-9 Essential Cell Biology, 4th ed. (© Garland Science 2014)

© 2016 Pearson Education, Inc.

Elongation of the RNA Chain

- Chain *elongation* continues as RNA polymerase moves along the DNA molecule
- The *RNA is elongated in the 5' to 3' direction,* with each new nucleotide added to the 3' end
- As the polymerase moves along the DNA strand, the double helix ahead of the polymerase is unwound (transcription bubble moves along), and the DNA behind it is rewound into a double helix

© 2016 Pearson Education, Inc.



RNA Pol resides on DNA at txn bubble

Synthesize mRNA (red) with template DNA (light blue)

Transient hydrogen binding btw template DNA and mRNA → RNA poly knows which nts to add

New nts are added at 3'site, and synthesized RNA exit from another site

Upstream DNA sequences that are already transcribed have already form double helix

© 2016 Pearson Education, Inc.

RNA Proofreading

- When an incorrect nucleotide is incorporated during transcription, the RNA polymerase backs up slightly and the few nucleotides with the incorrect nucleotide are removed, in a process called **RNA backtracking**
- This is RNA proofreading; occasional errors in RNA molecules are not as critical as errors in DNA replication. Single copy of DNA vs 100s/1000s copies of mRNA!
- Mutations that might have aroused in DNA sequence can't be corrected by RNA Pol.

© 2016 Pearson Education, Inc.

Types of Termination Signal

- How RNA polymerase knows where to stop? There needs to be a signal for transcription termination
- Either GC-rich hairpin loop formation or through rho factor.
- RNA molecules that don't form the GC-rich hairpin require the **rho factor** for termination
- The rho factor binds to specific termination sequences 50–90 bases long located near the 3' end of newly forming RNA molecules.
- The rho factor is an ATP-dependent unwinding enzyme moving along the RNA molecule toward the 3' end and unwinding it from the DNA template as it proceeds

© 2016 Pearson Education, Inc.

Eukaryotic Transcription

© 2016 Pearson Education, Inc.

Transcription in Eukaryotic Cells Has Additional Complexity Compared with Prokaryotes

- Eukaryotic transcription involves the same four stages as prokaryotic, but there are several important differences
  1. *Three different RNA polymerases* transcribe one or more different classes of RNA
  2. *Eukaryotic promoters* are more varied than bacterial ones; some are even located *downstream* of the gene
  3. RNA polymerases in eukaryotes require additional proteins called *transcription factors,* some of which must bind *before* the RNA polymerase can bind.
  4. Newly forming RNA molecules undergo *RNA processing,* chemical modification during and after transcription

© 2016 Pearson Education, Inc.

RNA Polymerase I, II, and III Carry Out Transcription in the Eukaryotic Nucleus

- There are three RNA polymerases in the nucleus, designated RNA polymerases I, II, and III
- These differ in their location in the nucleus and the types of RNA they synthesize.

Location for rRNA (ribosome) synthesis

Table 18-1 Properties of Eukaryotic RNA Polymerases

RNA Polymerase	Location	Main Products	α-Amanitin Sensitivity
I	Nucleolus	Precursor for 28S rRNA, 18S rRNA, and 5.8S rRNA	Resistant
II	Nucleoplasm	Pre-mRNA, most snRNA, and microRNA	Very sensitive
III	Nucleoplasm	Pre-tRNA, 5S rRNA, and other small RNAs	Moderately sensitive*
Mitochondrial	Mitochondrion	Mitochondrial RNA	Resistant
Chloroplast	Chloroplast	Chloroplast RNA	Resistant

*In mammals.

© 2016 Pearson Education, Inc.

Eukaryotic Transcription

- Similar to prokaryotic transcription but much more complex.
- Promoter (transcription start) regions contain specific DNA sequences that are targeted by transcription factors (TF), similar to sigma factor in prokaryotes.
- TATA box is bound by TATA-binding protein (TBP), a TF.
- Other transcription initiation factors get recruited to promoters, along with RNA polymerase.
- Upon melting of DNA double helix, all TFs are lost and RNA polymerase starts transcribing.
- Monocistronic mRNA

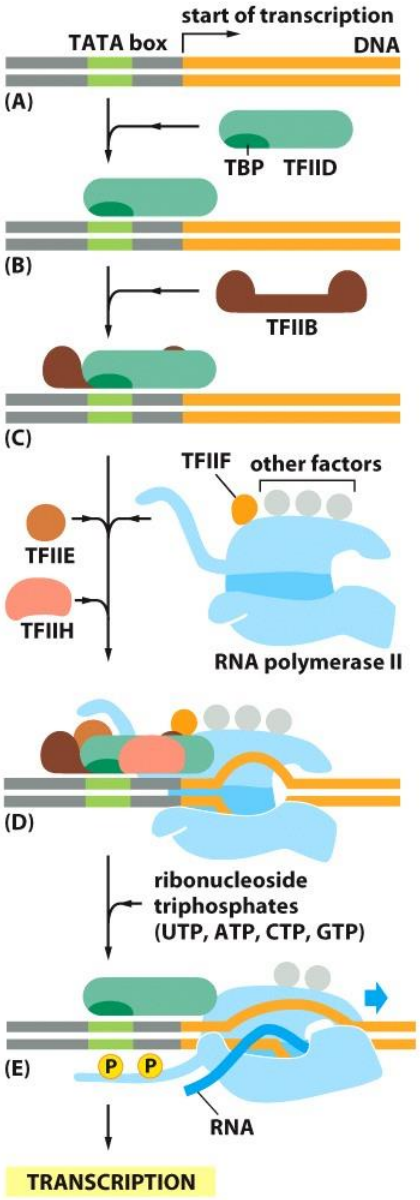


Figure 7-12 Essential Cell Biology, 4th ed. (© Garland Science 2014)

© 2016 Pearson Education, Inc.

Additional Control Elements

- Core promoters are capable of driving only a *basal* (low) level of transcription
- Additional short sequences upstream (*upstream control elements=ENHANCER*) improve the promoter's efficiency.

© 2016 Pearson Education, Inc.

Transcription is mediated by long distance sequences (enhancers) and numerous proteins that bind promoters and enhancers.

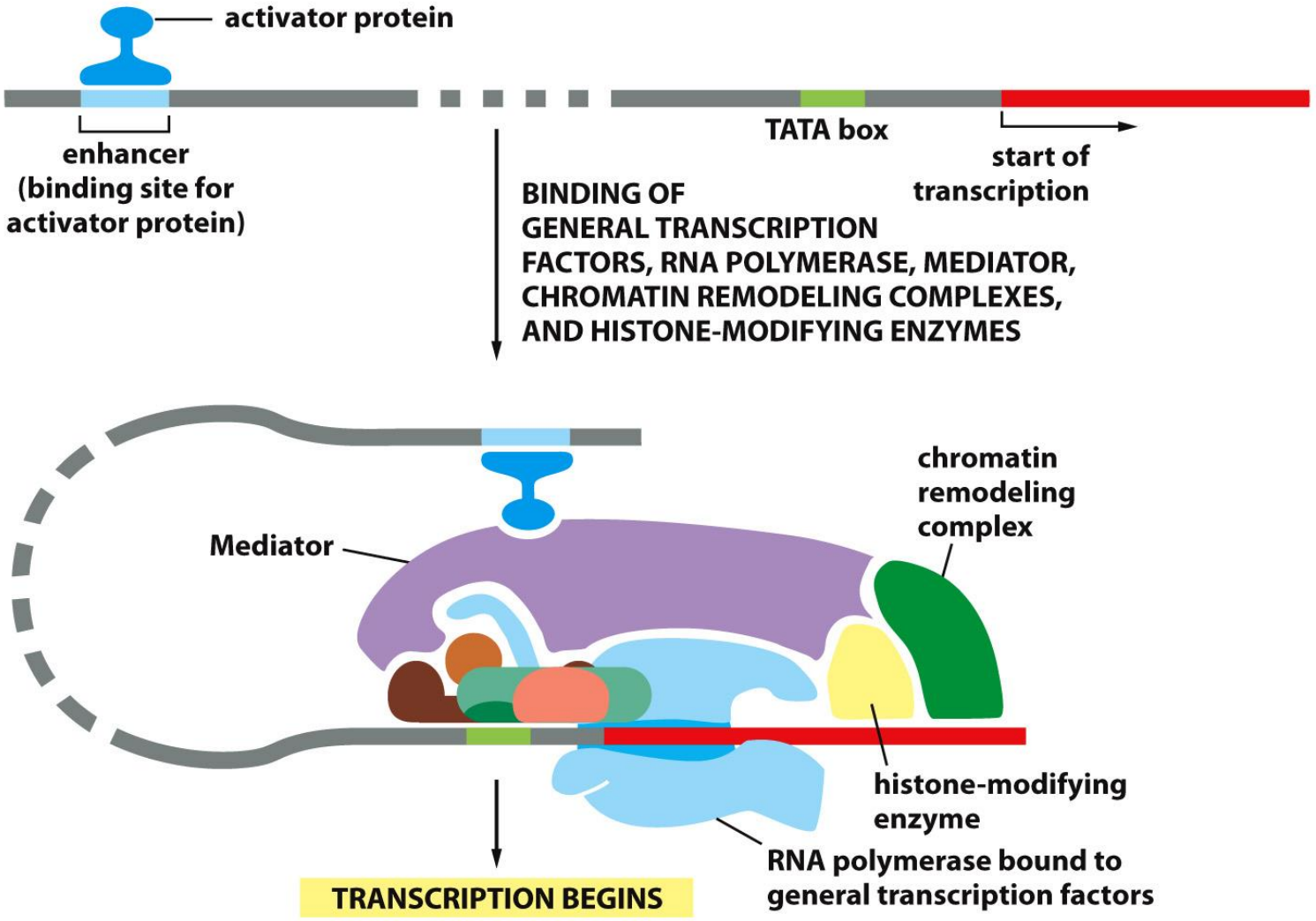


Figure 6-18 Molecular Biology of the Cell 6e (© Garland Science 2015)

Other Proteins Needed for Transcription

- Besides general transcription factors and RNA polymerase II, several other kinds of proteins are needed
- Some open chromatin structure to make DNA accessible to RNA polymerase; others are *regulatory factors* that bind upstream control elements and recruit *coactivator proteins*
- Mediate cell type-specific or time-specific txn.

© 2016 Pearson Education, Inc.

Elongation, Termination, and RNA Cleavage Are Involved in Completing Eukaryotic RNA Synthesis

- After initiation, RNA polymerases move along the DNA and synthesize a complementary RNA
- Termination is governed by signals that differ for each type of RNA polymerase
- For RNA polymerase II, transcripts are cleaved at a specific site before transcription ceases
- The cleavage site is 10–35 nucleotides downstream of a AAUAAA sequence in the RNA

© 2016 Pearson Education, Inc.

Polyadenylation

- The cleavage site of polymerase II transcripts is also the site for addition of a *poly*(*A*) *tail*
- This is a string of adenine nucleotides added to the 3' end of most eukaryotic mRNAs

© 2016 Pearson Education, Inc.

- RNAs perform many different functions within cells. Thus far, you have learned about transcription primarily in the context of production of mRNAs, RNAs that usually encode proteins. However, mRNAs are only one of many RNAs found in cells.

Table 18-2 Classes of Naturally Occurring RNA

Type of RNA	Function(s)
I. Messenger RNA	Provides information for production of protein
II. RNAs involved in translation	
Transfer RNAs (tRNAs)	Bring the correct amino acid to mRNA during translation
Ribosomal RNAs (rRNAs)	Major components of ribosomes; guide and catalyze assembly of polypeptides from mRNA during translation
III. RNAs involved in RNA processing	
Small nuclear RNAs (snRNAs)	Form components of the spliceosome; involved in splicing of eukaryotic RNAs
Small nucleolar RNAs (snoRNAs)	Processing of rRNAs in nucleolus
IV. RNAs involved in regulation of gene expression and other events	
MicroRNAs (miRNAs)	Regulate stability and translation of mRNAs
Small interfering RNAs (siRNAs)*	Inhibit production of viruses, suppress spread of transposable elements in plants
Piwi-interacting RNAs (piRNAs)	Suppress spread of transposable elements in germ cells in animals
Long noncoding RNAs (lncRNAs)	Some regulate chromatin; others may regulate transcription or RNA processing

*Not to be confused with small inhibitory RNAs, which are synthetic RNAs used in gene knockdown experiments (see Chapter 21).

© 2016 Pearson Education, Inc.

DNA Transcription (Basic)

https://www.youtube.com/watch?v=8M198uHJd_8

DNA Transcription (Advanced)

https://www.youtube.com/watch?v=-AnsJlLjbz8

© 2016 Pearson Education, Inc.

RNA Processing

mRNA Processing in Eukaryotes

© 2016 Pearson Education, Inc.

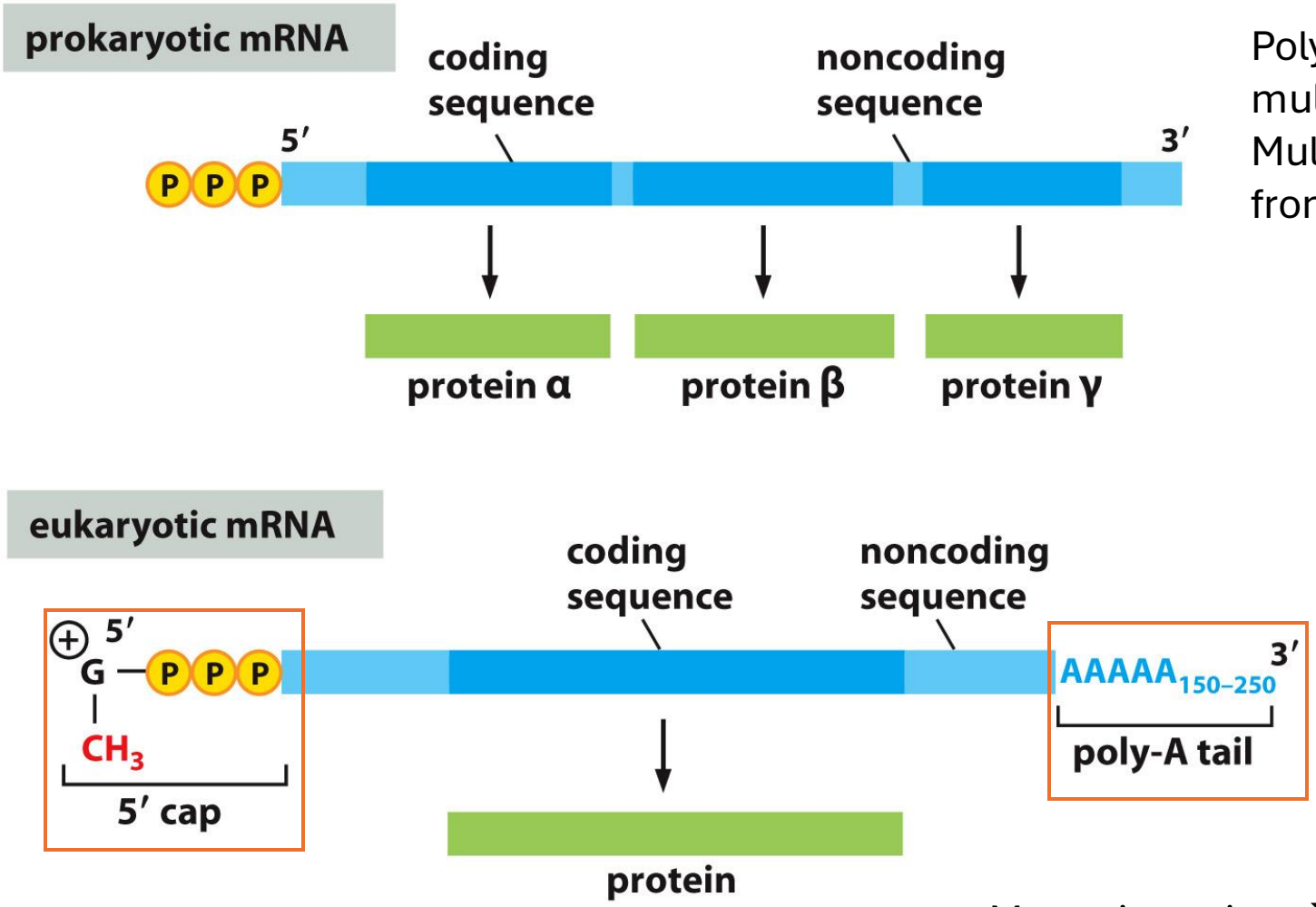


Figure 6-21a Molecular Biology of the Cell 6e (© Garland Science 2015)

Polycistronic → multiple coding units. Multiple polypeptides from single mRNA

Monocistronic → single coding unit

Code for a single polypeptide from each mRNA

© 2016 Pearson Education, Inc.

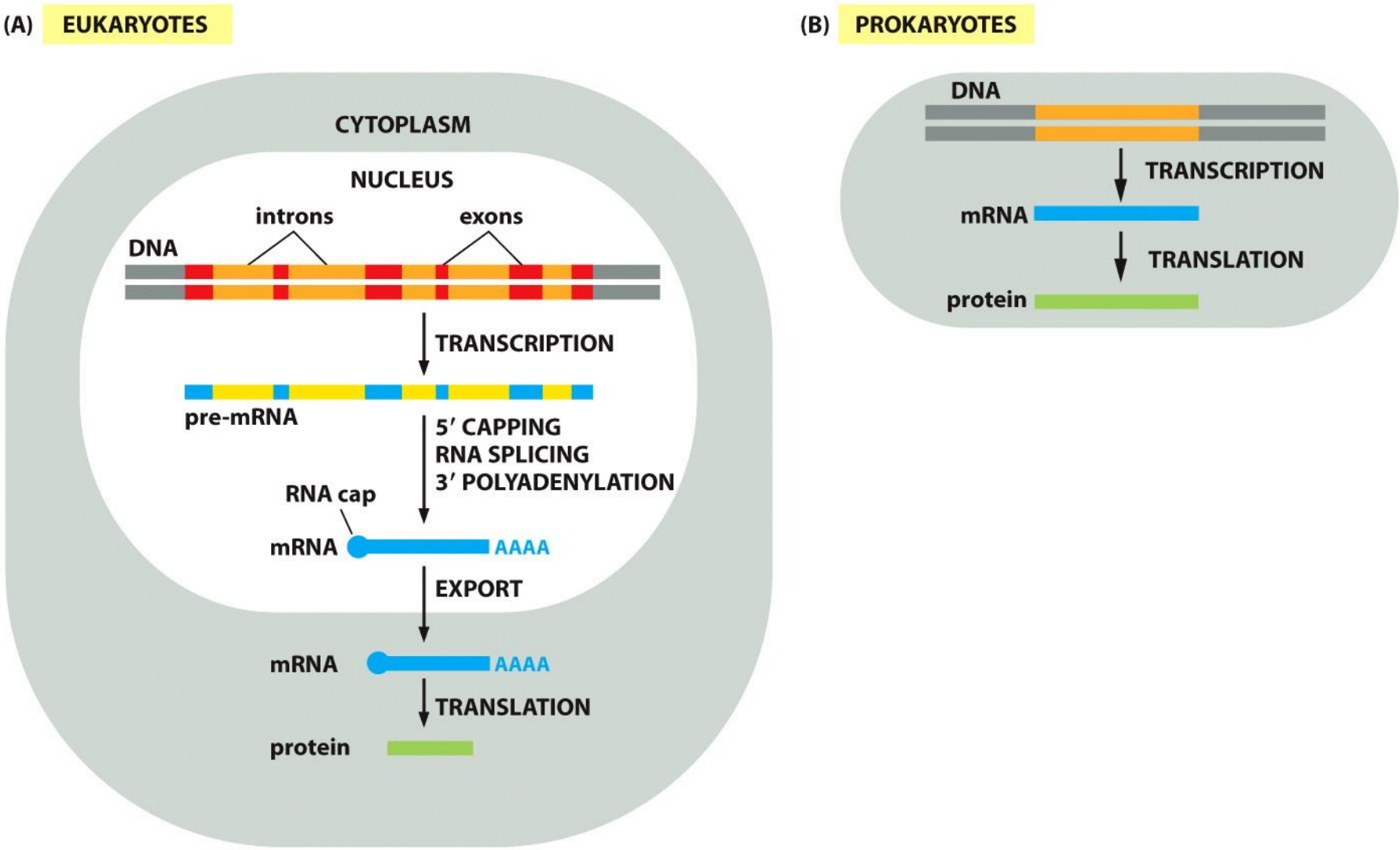


Figure 7-24 Essential Cell Biology, 4th ed. (© Garland Science 2014)

© 2016 Pearson Education, Inc.

Messenger RNA Processing in Eukaryotes Involves Capping, Addition of Poly(A), and Removal of Introns

- Most bacterial RNA is synthesized in a form that is ready for translation with no need for processing
- Because there is no nuclear membrane, bacterial transcripts are translated as they are transcribed

© 2016 Pearson Education, Inc.

Transcription and Translation in Eukaryotes

- Eukaryotic transcripts must be exported from the nucleus to be translated
- Substantial processing occurs in the nucleus before export
- Primary transcripts are often very long, 2000–20,000 nucleotides (**pre-mRNA**).
- Pre-mRNAs are processed by removal of sequences and addition of 5' caps and 3' tails
- The C-terminal domain of one of the subunits of RNA polymerase II acts as a platform for protein complexes involved in processing

© 2016 Pearson Education, Inc.

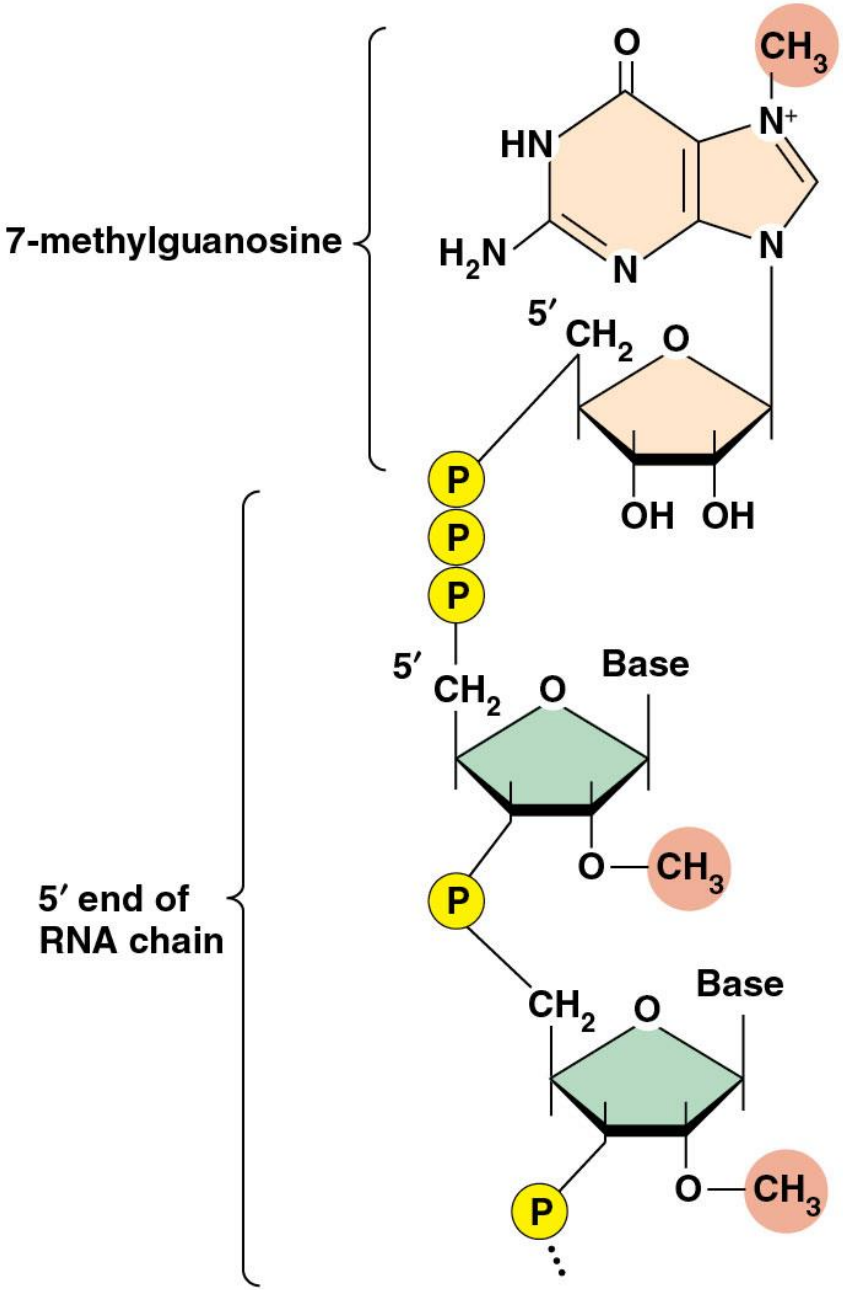
- **mRNA processing in eukaryotes:**
  - 5’ cap
  - 3’ polyA tail
  - Splicing (removal of introns)

© 2016 Pearson Education, Inc.

5' Caps and 3' Poly(A) Tails

- Eukaryotic mRNAs have a modified nucleotide called the **5' cap,** and the 3' ends have a long stretch of adenines called the **poly(A) tail**
- The 5' cap is a guanosine that is methylated at position 7 of the purine ring
- It is bound to the RNA molecule by a 5'→5' linkage rather than the usual 3'→5' bond

© 2016 Pearson Education, Inc.



The 5' cap is a methylated guanosine. It is bound to the RNA molecule by a **5'→5' linkage** rather than the usual **3'→5'** bond (covalent bond between nucleotides

© 2016 Pearson Education, Inc.

Roles of the 5' Cap

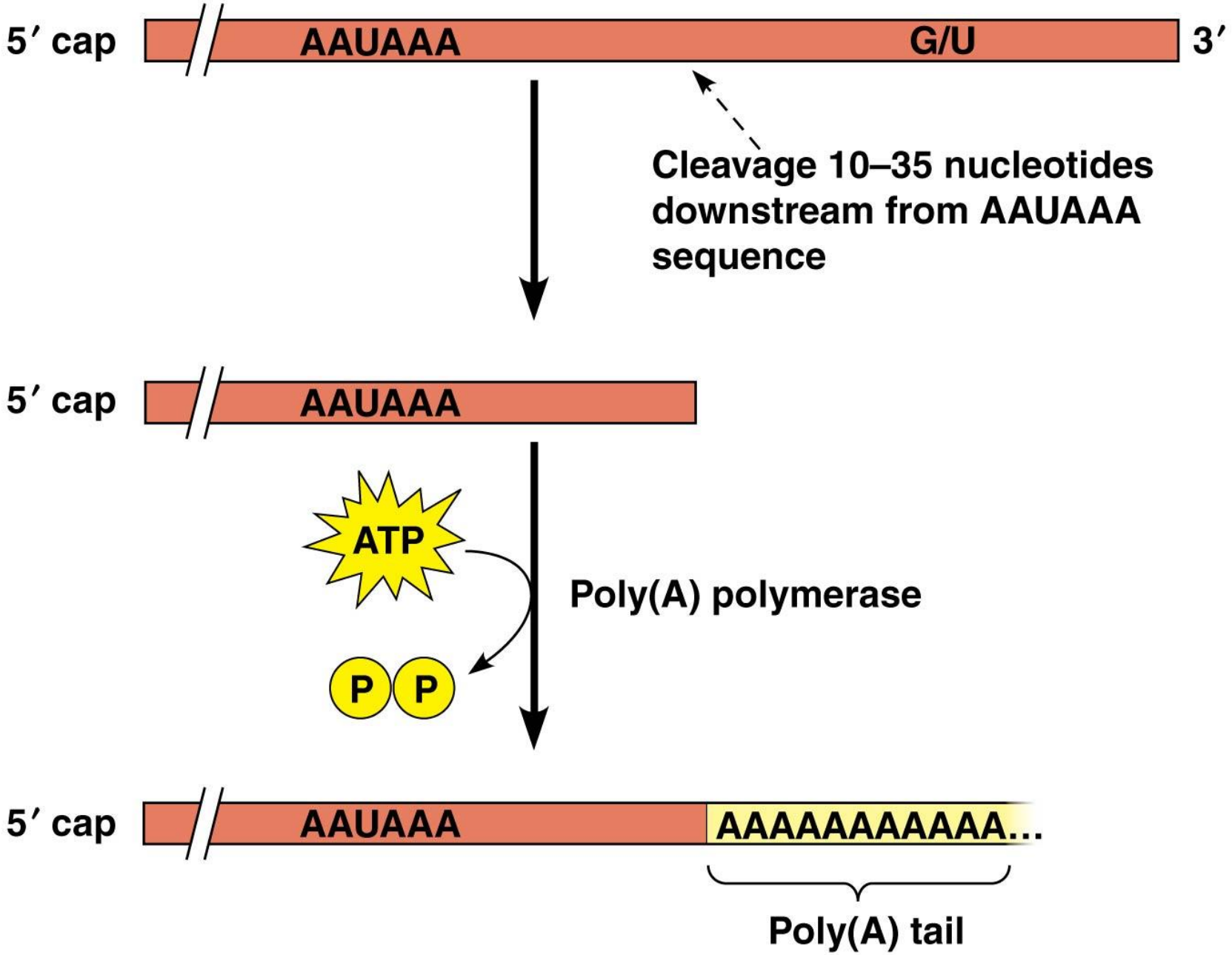
- The 5' cap is added soon after transcription is initiated.
- The cap contributes to **mRNA stability** by protecting the RNA from nucleases
  - Rnases are specialized to cut from covalent bonds (3'→5'), they don't recognize **5'→5' linkage**
- The cap also plays a role in positioning the RNA on the ribosome for initiation of translation

© 2016 Pearson Education, Inc.

The Poly(A) Tail

- The poly(A) tail ranges from 50 to 250 nucleotides long and is added by the enzyme *poly(A) polymerase*
  - Genome sequence doesn't have many Ts at the end, Poly(A) tail is not coded by RNA polymerase, added later by another enzyme.
- A signal for addition of the poly(A) tail, AAUAAA, is located just upstream of the polyadenylation site, and a GU- or U-rich element is located downstream of it

© 2016 Pearson Education, Inc.



© 2016 Pearson Education, Inc.

Function of the Poly(A) Tail

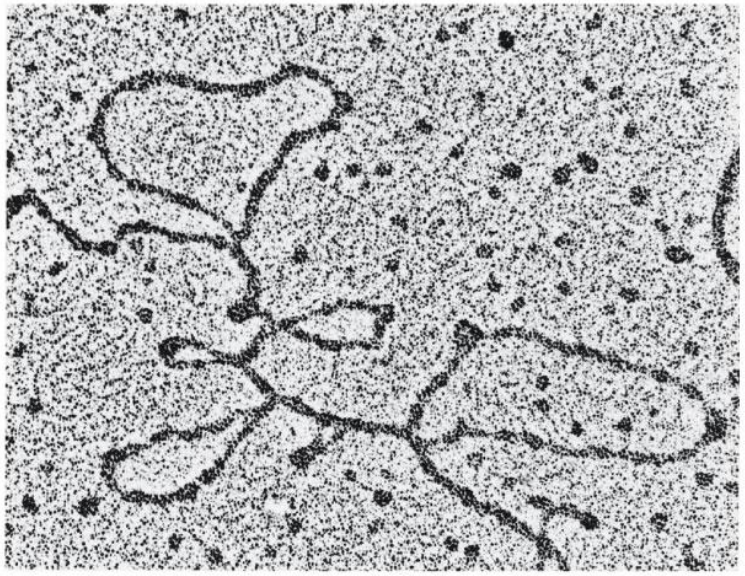
- The poly(A) tail protects the mRNA from nuclease attack; the length of the tail influences stability
- It is also required for export of the transcript to the cytoplasm
- It may also help ribosomes recognize and bind mRNAs
- Function only in eukaryotic mRNAs, not in prokaryotic mRNAs or eukaryotic non-coding RNAs.

© 2016 Pearson Education, Inc.

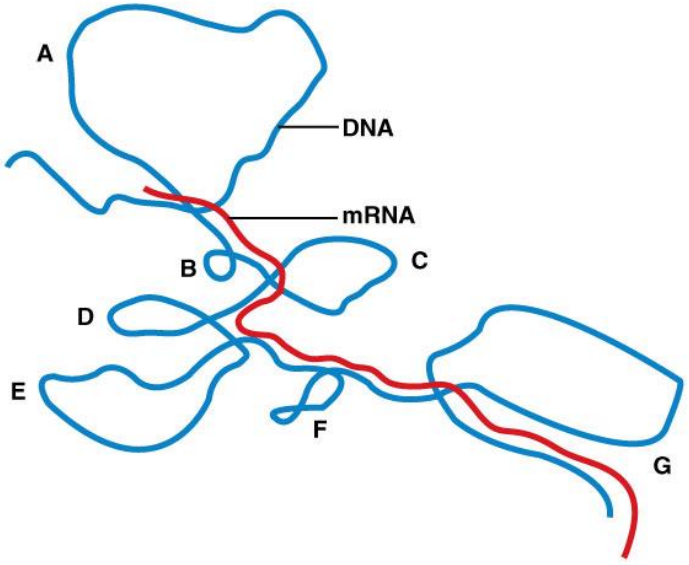
R Looping Observation

- Scientist performed an experiment: Single-stranded RNA and double-stranded DNA are hybridized under conditions that favor *heteroduplexes,* RNA-DNA hybrid molecules.
- DNA that is not hybridized to RNA is displaced as a single-stranded loop, easily observed with electron microscopy
- Some genes contained *multiple loops*

© 2016 Pearson Education, Inc.



(a)



(b)

R Loops: DNA sequences that are not found in mature mRNAs, they are not complementary.

© 2016 Pearson Education, Inc.

Exons and Introns

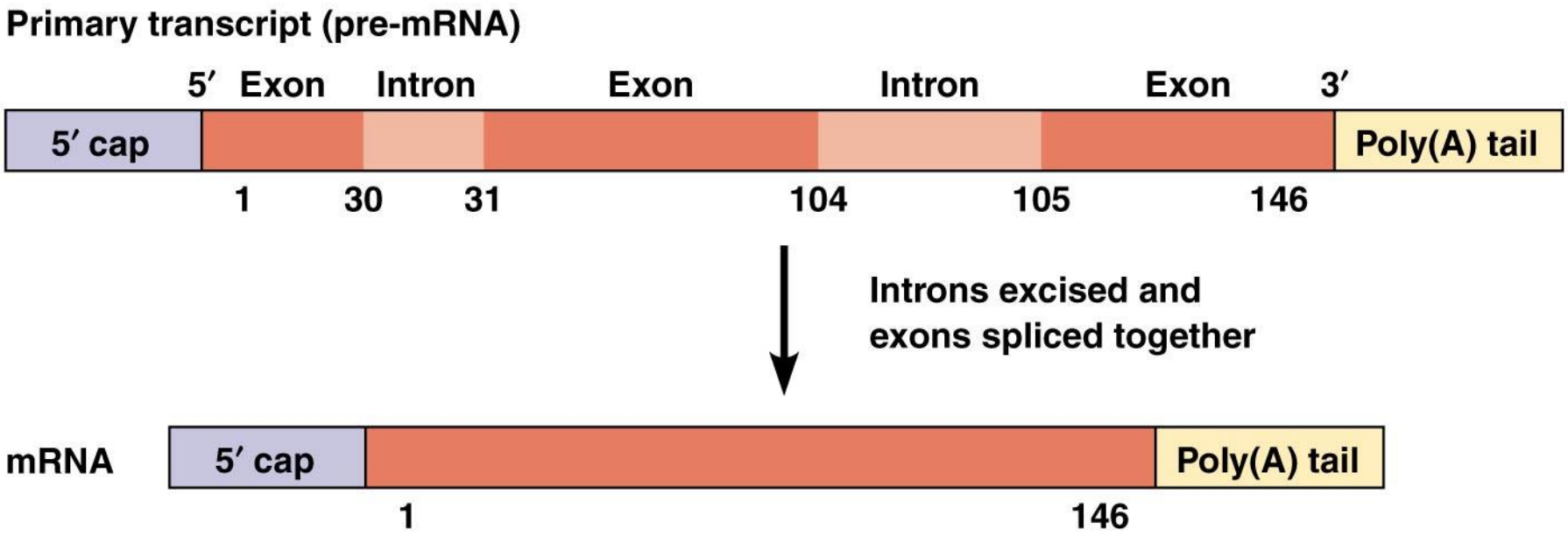
- Sequences that appear in the final mRNA are called **exons**
- Introns are present in most protein-coding genes of multicellular eukaryotes
- The size and number of introns varies considerably
- In rare cases, introns are processed to yield functional products
- **However, most introns are destroyed without serving any obvious function**

© 2016 Pearson Education, Inc.

Spliceosomes Remove Introns from Pre-mRNA

- The process of removing introns and joining the exons is **RNA splicing**
- About 15% of inherited human diseases involve splicing errors; such errors lead to incorrect protein products
- Sequences commonly found at the intron-exon boundaries likely determine the *5' and 3' splice sites*

© 2016 Pearson Education, Inc.



© 2016 Pearson Education, Inc.

No splicing!
Smal genome size

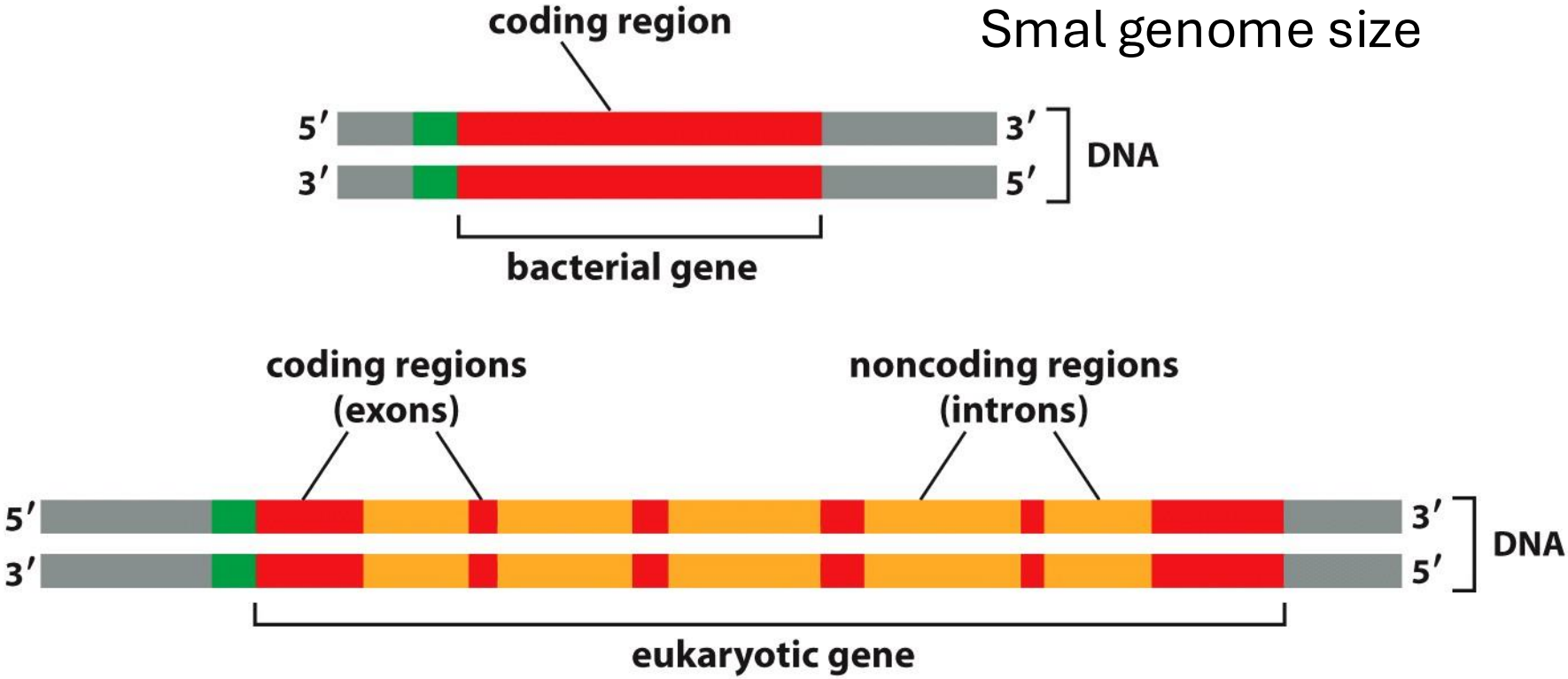


Figure 7-17 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Introns are transcribed, but not found in mature mRNA. The length of genes increase quite considerably.

© 2016 Pearson Education, Inc.

3 exons,
4
introns
in small
gene

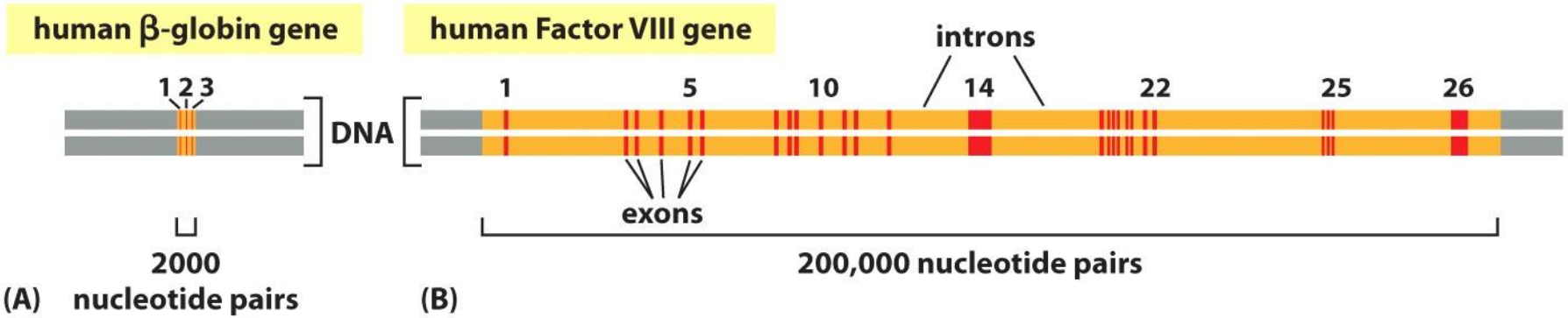


Figure 7-18 Essential Cell Biology, 4th ed. (© Garland Science 2014)

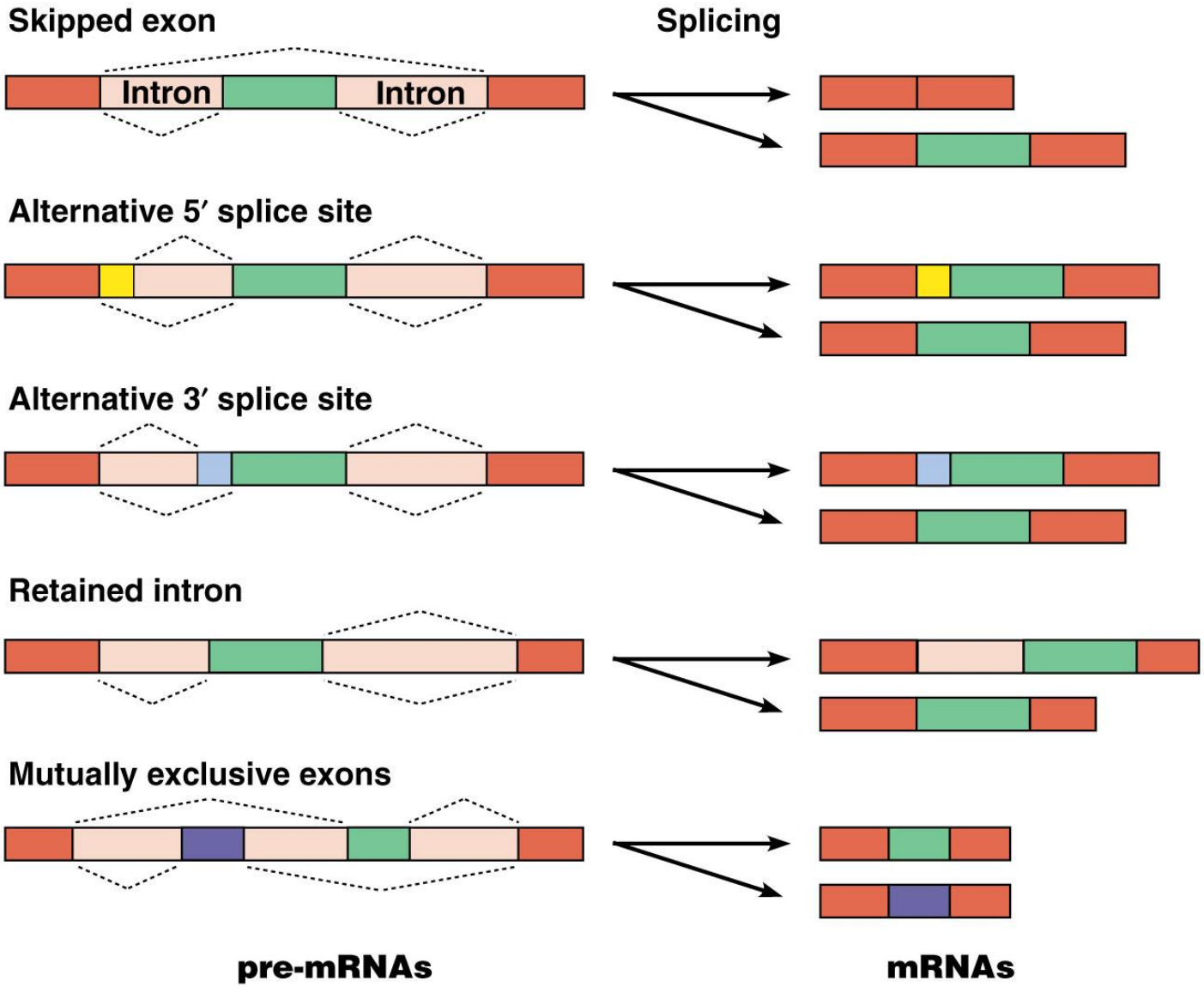
A lot of exons and introns with variable length.

© 2016 Pearson Education, Inc.

Alternative Splicing

- The presence of introns allows each gene's pre-mRNA molecule to be spliced in multiple ways, leading to production of multiple protein products
- This **alternative splicing** is possible via mechanisms allowing certain splice sites to be activated or skipped
- Different function of the same polypeptide can be achieved from a single transcription event.

© 2016 Pearson Education, Inc.



Control over these splice sites is exerted by various molecules, including regulatory proteins and RNAs, that bind to splicing enhancer or splicing silencer sequences in pre-mRNAs.

© 2016 Pearson Education, Inc.

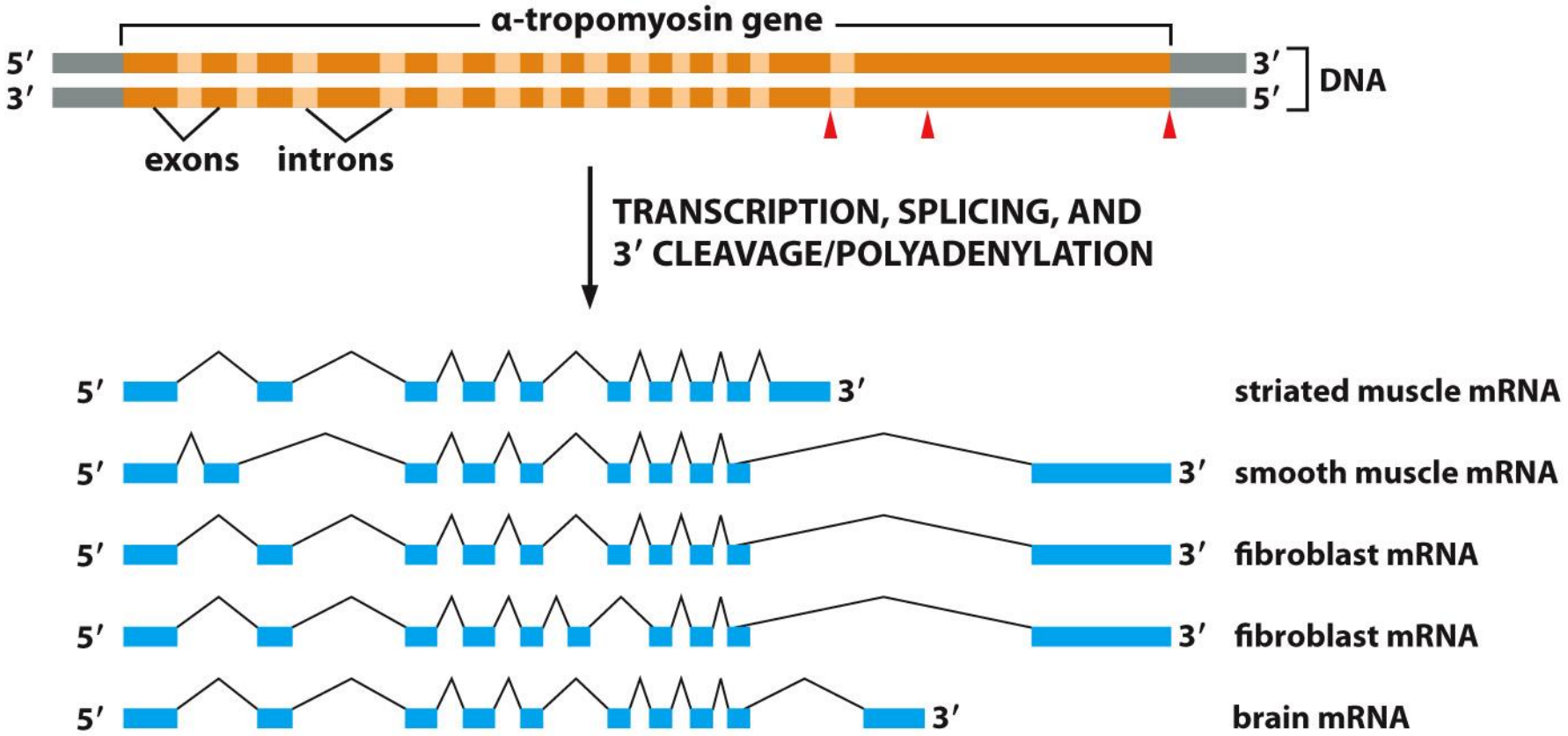


Figure 6-26 Molecular Biology of the Cell 6e (© Garland Science 2015)

Instead of increasing the number of genes, humans transcribe the majority of their genes into pre-mRNAs that can be spliced in more than one way. As a result, the roughly 25,000 human genes produce mRNAs encoding hundreds of thousands of different polypeptides.

© 2016 Pearson Education, Inc.

Intron Functions

- Besides alternative splicing, introns allow the evolution of new protein-coding genes through recombination events
- Recombination between introns produces new combinations of exons—*exon shuffling*
- It can also produce duplicate copies of exons within a gene, one of which could mutate to a new sequence

© 2016 Pearson Education, Inc.

Most mRNA Molecules Have a Relatively Short Life Span

- Most mRNA molecules have a high *turnover rate* (rate at which molecules are degraded and replaced).
- In this respect, mRNA contrasts with the other major forms of RNA in the cell, rRNA and tRNA, which are notable for their stability.
- Turnover is measured in terms of *half-life*, the time required for 50% of the molecules to degrade → affects amount of protein to be translated
- mRNA molecules of eukaryotes have half-lives of several hours to a few days; in bacteria, the half-lives are usually only a few minutes

© 2016 Pearson Education, Inc.

The Abundance of mRNA Allows Amplification of Genetic Information

- mRNA can be synthesized again and again from a piece of template DNA, providing an opportunity for *amplification* of genetic information
- For example, the haploid genome of the silkworm has only one copy of the fibroin gene, but about 10^4 copies of the mRNA are transcribed from two copies of the gene

© 2016 Pearson Education, Inc.